FEDERAL URDU UNIVERSITY OF ARTS, SCIENCE & TECHNOLOGY

Graduate Research Management System (GRMS)

Streamlining Academic Research Workflows with Precision

UNIVERSITY & DEPT.
FUUAST | Computer Science

SEMESTER & SESSION
BSCS [2022 - 2026]

PROJECT SUPERVISOR
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PRESENTATION AGENDA

PROJECT INTRODUCTION

The Graduate Research Management System (GRMS) is a dedicated web-based platform designed to automate and digitalize the research lifecycle at FUUAST.
It centralizes academic activities from student admission to final degree issuance, ensuring transparency and data integrity for postgraduate and undergraduate research governance.

PROBLEM STATEMENT

Manual Handling

Current research processes are managed through basic spreadsheets and physical files, leading to data fragmentation.

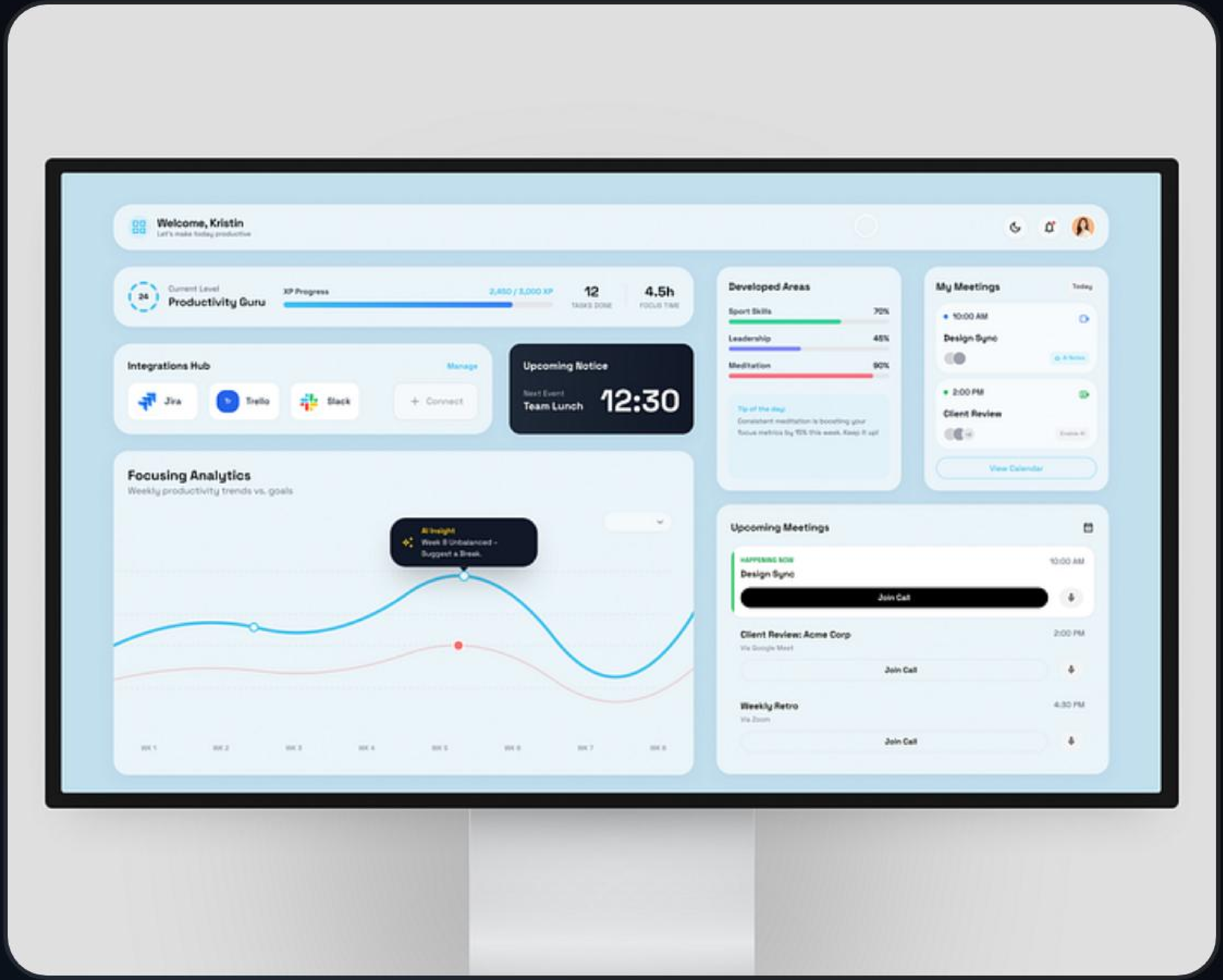
Communication Gaps

Significant delays in communication between students, supervisors, and the GRMC committee.

Record Tracking

Difficulty in maintaining and tracking historical research records and extension case timelines.

PROJECT SCOPE



- **End-to-End Governance:** Digital handling of admissions, synopsis, and thesis evaluation.
- **Role-Based Access:** Dedicated dashboards for Admin, Student, Supervisor, and Examiner.
- **Document Engine:** Automated PDF generation for degree letters and reports.
- **Scalability:** Designed for multi-department and multi-campus environments.

CORE MANAGEMENT MODULES

Admission & Enrollment

Handles student registration, status management, and supervisor assignment.

Synopsis Workflow

Submission, review, and multi-stage approval tracking for research proposals.

Thesis Evaluation

Examiner assignment, report collection, and viva result management.

Meeting Records

Scheduling and logging of student-supervisor engagement points.

Extension Tracking

Automated flagging of research timelines and approval of case extensions.

Degree Generation

Verification of completion and automated issuance of official degree letters.

TECHNOLOGY ARCHITECTURE

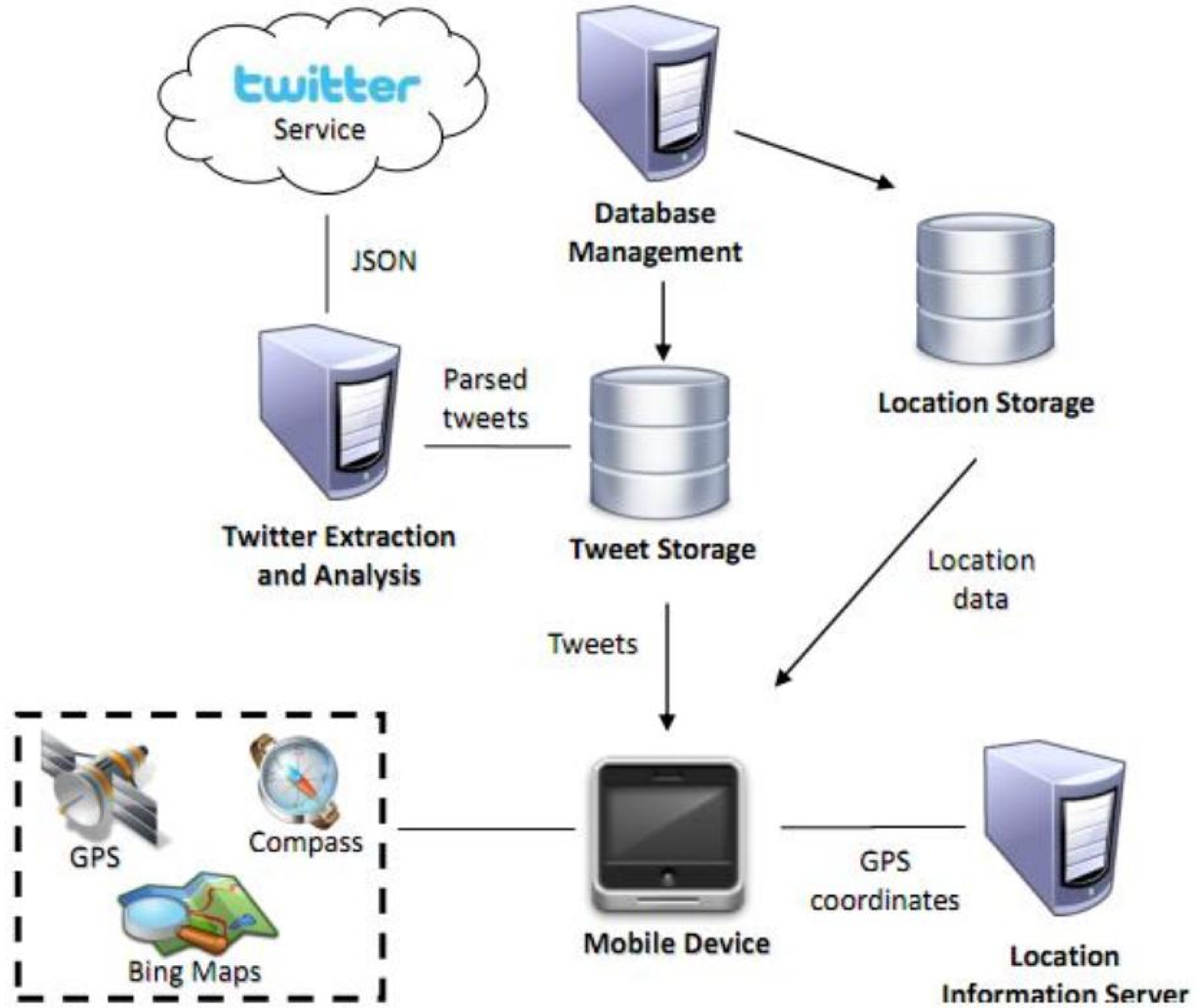


Figure 2

Layer	Technology Used
Frontend	HTML5, CSS3, JavaScript, Bootstrap 5
Backend	Django (Python Web Framework)
Database	MySQL (Normalized 3NF)
Deployment	Railway Architecture

IMPLEMENTATION STRATEGY

- **Modular Architecture:** Decoupled layers for business logic and presentation.
- **Security Matrix:** Cryptographic hashing for passwords and role-based validation.
- **Iterative Model:** Agile development cycles with continuous testing.
- **CI/CD Pipeline:** Automated build and deployment via GitHub webhooks.

Production App

`grms-production.up.railway.app`

Live production environment for institutional access.

PROJECT ARTIFACTS

Project Proposal

Comprehensive roadmap and research objectives.

Access Document

SRS Document

Functional and Non-Functional requirement specifications.

Access Document

User Manual

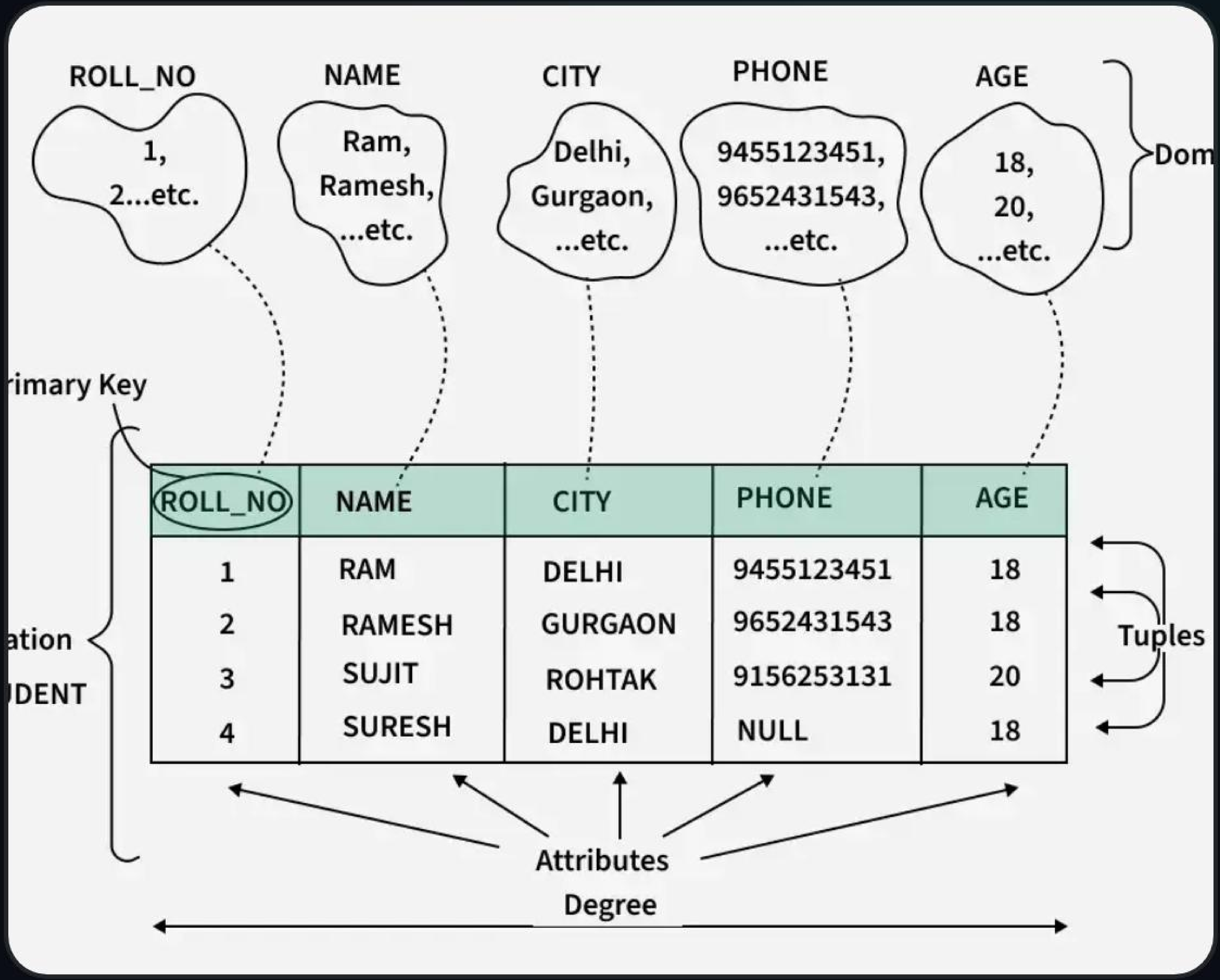
Step-by-step guide for Students, Supervisors, and Admins.

Access Document

* All artifacts are formatted to institutional standards for the Award of BSCS Degree.

DESIGN & CODING STANDARDS

- **Pythonic Code:** Adherence to PEP 8 standards for readability.
- **Database Normalization:** 3NF schemas to eliminate data redundancy.
- **Responsive UI:** Mobile-first design approach using Bootstrap.
- **Authentication:** Django-integrated secure session management.



SYSTEM TESTING & MILESTONES

Testing Categories

- **Unit Testing:** Validation of individual logic components.
- **Integration Testing:** Cross-module workflow verification.
- **UAT:** User acceptance trials with pilot student data.

Success Metric: 100% pass rate on core functional test cases (Admission, Synopsis, Thesis).

PROJECT RESOURCES

GitHub Repo

Access the source code and development history here.

Open Repository

Project Docs

Complete documentation and research specifications.

Read Docs

Live App

View the production deployment of the system.

Visit Site

THANK YOU!

Any Questions Regarding GRMS?

Developed By: Alishba Arshad & Sana Tariq

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grms-production.up.railway.app

IMAGE SOURCES

https://lmnarchitects.com/wp-content/uploads/2024/08/Undergraduate-Academic-Building-UC-Berkeley_Image-Page.jpg

Source: lmnarchitects.com

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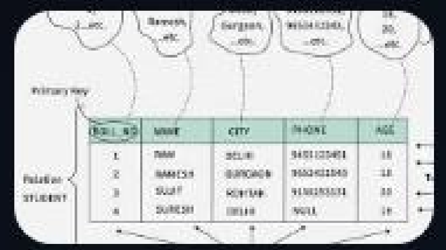

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